

Predicting NBA Fantasy Points

Short stakeholder report for the final machine-learning project

Prepared for lecturer review | Final run folder: runs/final

Final test MAE

7.671

2025 unseen season

Final test R2

0.544

2025 unseen season

Model features

11

lineup-aware CatBoost

This project predicts one NBA player-game fantasy score before the game outcome is known. The final report emphasizes a strict time-aware workflow: historical training seasons, a later validation season, and a single untouched test season. The main modeling risk was leakage from current-game minutes and box-score outcomes. That risk was explicitly diagnosed, separated from valid models, and excluded from the final recommendation.

Final recommendation

Use the catboost model with the lineup_aware_starter_flag_roll10 feature set when confirmed starters are available. It delivered validation MAE 7.644 and final 2025 test MAE 7.671, while staying interpretable and leakage-controlled.

Important distinction

The lowest valid validation experiment was Best Isolation Forest anomaly model with MAE 7.580. It was kept as an advanced experiment because PCA/Isolation features were less interpretable than the selected 11-feature controlled model.

Data, Target, And Prediction Moment

What was predicted, from which rows, and how the evaluation was separated over time.

Prediction unit: one row represents one player in one NBA regular-season game. The target is calculated after the game, but model features must be available before the prediction moment.

Fantasy points formula

fantasy_points = points + 1.2*rebounds + 1.5*assists + 3*steals + 3*blocks - turnovers

split	rows	unique_games	unique_players	seasons	target_mean	target_std
train	76937	3624	818	2021, 2022, 2023	21.46	14.87
validation	26162	1223	569	2024	21.72	14.91
test	26651	1230	582	2025	21.62	14.50

Modeling rows

129,750

player-games

Modeling columns

576

after feature engineering

Test use

Once

after model selection

- ? Data source: Kaggle historical NBA data and player box scores downloaded by the final notebook through kagglehub.
- ? Train seasons were 2021-2023, validation was the full 2024 season, and test was the full 2025 season.
- ? The chronological split matches the real use case: predicting future games from past information.

Process And Leakage Control

The main quality issue was preventing future or same-game information from entering model inputs.

1. Data ingestion

Download raw NBA tables; inventory and filter regular-season rows.

2. Table unification

Create one player-game source table with player, team, opponent, starter, and target fields.

3. Feature engineering

Build shifted rolling averages and volatility features, so each row only uses previous games.

4. Split discipline

Train on 2021-2023, validate on 2024, and reserve 2025 as untouched test.

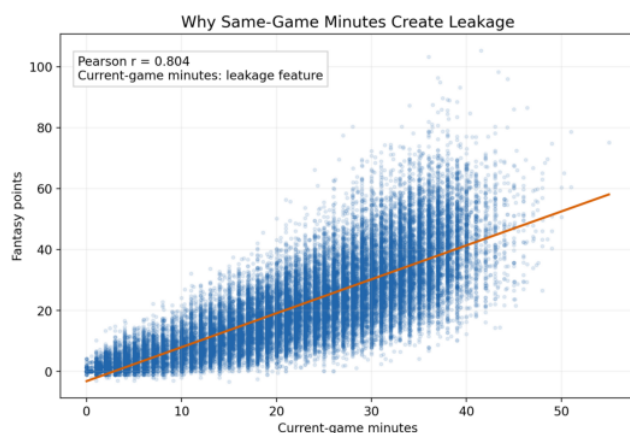
5. Modeling

Compare baselines, linear models, tree ensembles, boosting, PCA, and anomaly features.

6. Reporting

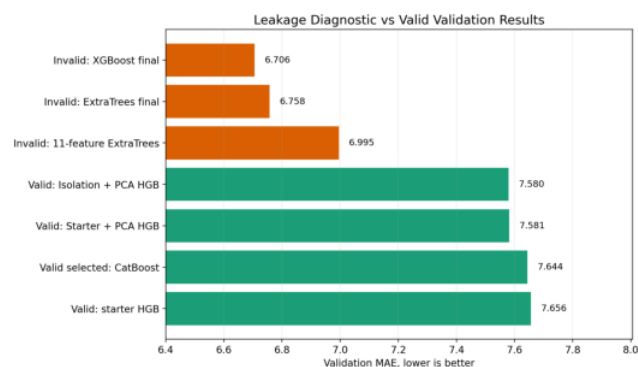
Summarize model performance, leakage diagnostics, feature importance, and limitations.

Leakage evidence



Current-game minutes had Pearson $r = 0.804$ with fantasy points. Direct minutes are therefore invalid as pre-game model inputs.

Invalid vs valid results

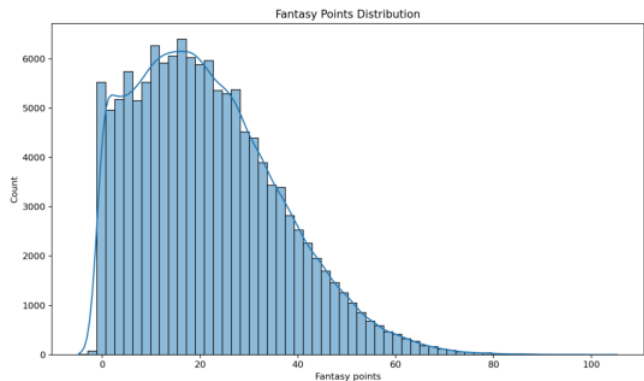


Orange bars are leakage diagnostics; green bars are valid validation results that exclude same-game minutes.

Exploratory Findings

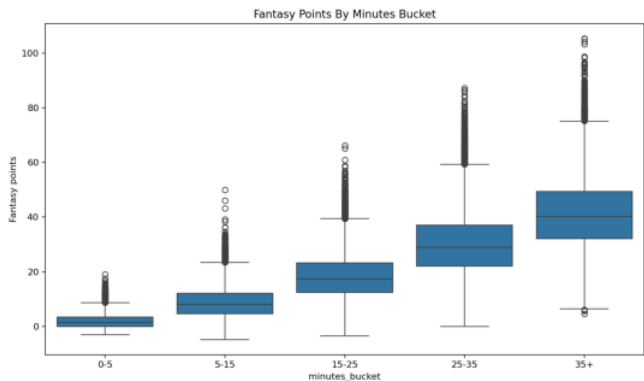
The useful signals are recent production, role, and opportunity; unstable features tend to be weaker.

Target distribution



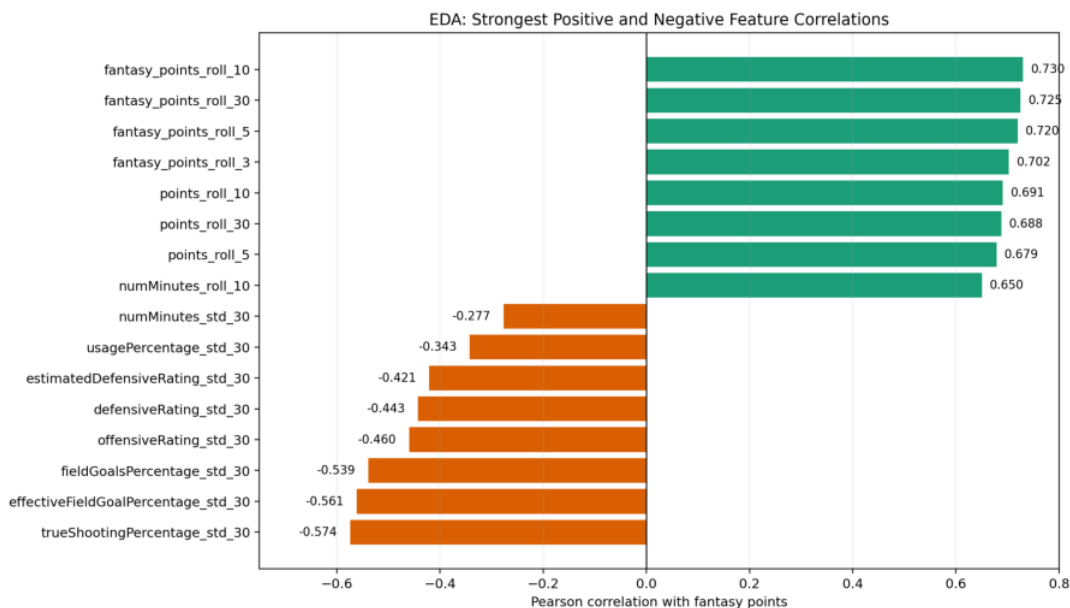
Fantasy points are right-skewed: many low or moderate scores and fewer very high games.

Minutes and opportunity



More minutes usually mean more opportunity, but same-game minutes are not available before the game.

Feature-target correlations



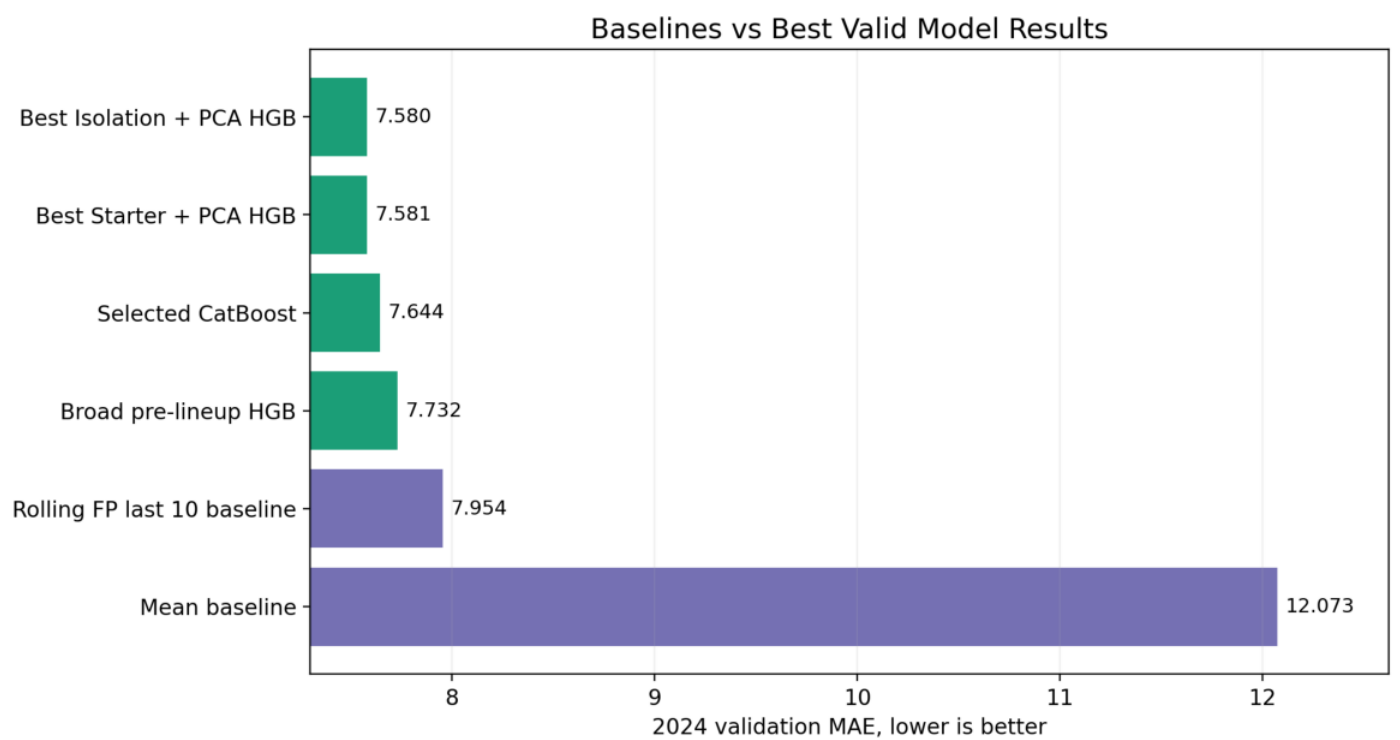
EDA takeaway

The data supports a basketball-intuitive model: predict from recent fantasy production, role, recent minutes, usage, and starter information. However, any current-game outcome column must be treated as invalid.

Model Comparison

Models were compared against simple baselines and against each other on validation performance.

Baselines vs valid models



The rolling last-10 fantasy baseline was strong, but the final models improved beyond it.

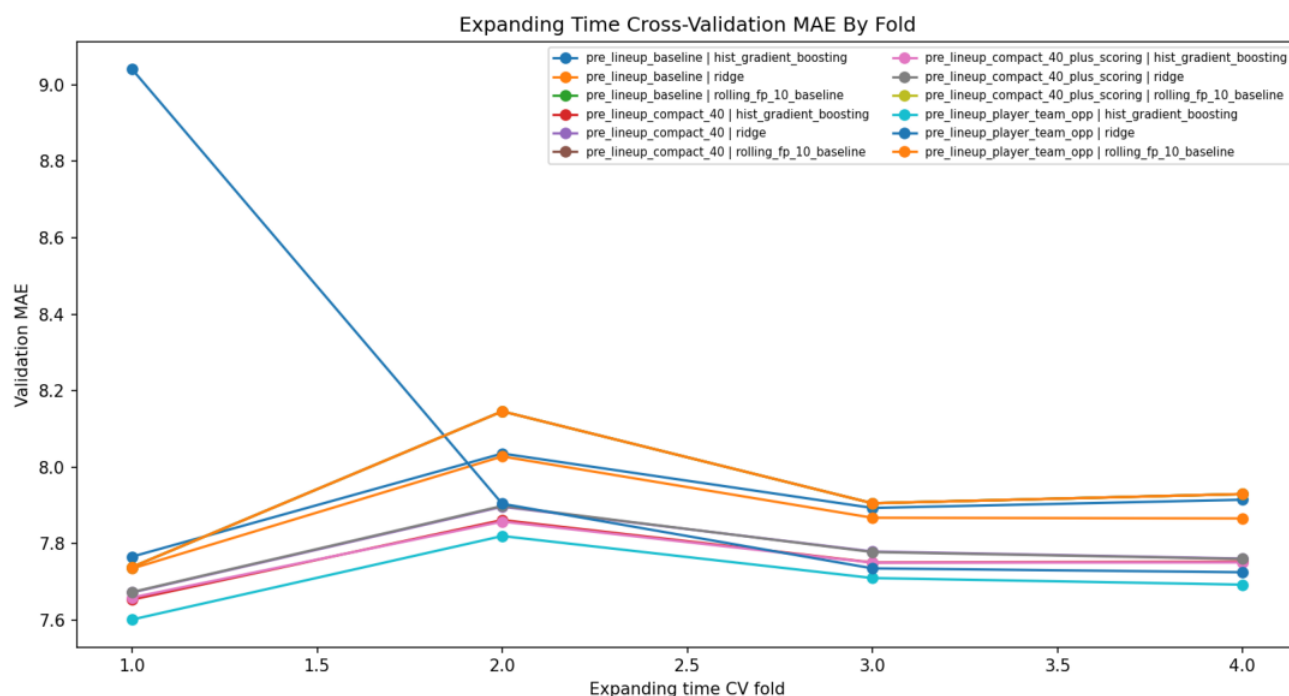
Model/result	Validity	Features	Val MAE	Interpretation
Mean baseline	baseline	45	12.073	Sanity check only
Rolling FP last 10	baseline	45	7.986	Strong simple benchmark
Broad pre-lineup HGB	pre-lineup	484	7.732	Many legal historical features
Selected CatBoost	lineup-aware	11	7.644	Final controlled model
Best Starter + PCA HGB	lineup-aware	41	7.581	Best PCA compression result
Best Isolation + PCA HGB	lineup-aware	44	7.580	Lowest valid validation MAE

- ? Linear models were useful controls but could not capture all nonlinear role/opportunity interactions.
- ? Boosting models performed best for this tabular regression problem.
- ? PCA and Isolation Forest improved validation MAE, but they add complexity and reduce interpretability.

Validation Discipline And Overfitting Checks

The project used both official validation and expanding time cross-validation.

Expanding time cross-validation



CV used earlier chronological blocks inside the training period; it did not train the final model and did not touch the 2025 test split.

feature_set	model	folds	mean_mae	std_mae	mae_range
player team opp	hist_gradient_boosting	4	7.706	0.090	0.218
compact 40 plus scoring	hist_gradient_boosting	4	7.755	0.081	0.199
compact 40	hist_gradient_boosting	4	7.755	0.085	0.208
compact 40	ridge	4	7.778	0.092	0.224
compact 40 plus scoring	ridge	4	7.778	0.093	0.225
baseline	ridge	4	7.874	0.120	0.293

Selected train MAE

7.614

2021-2023

Selected val MAE

7.644

2024

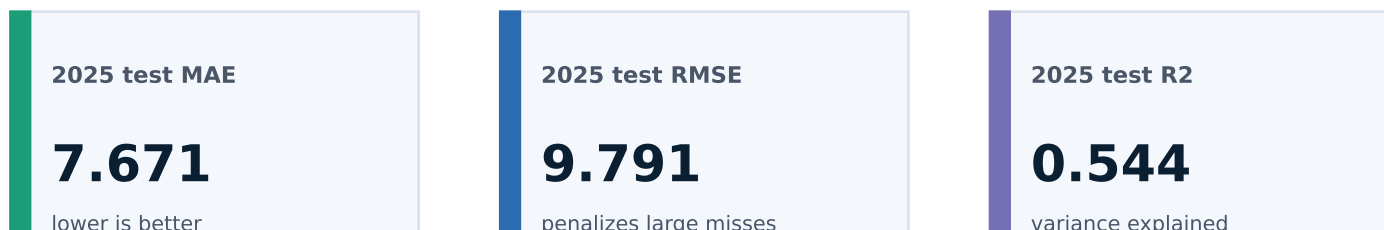
MAE gap

0.030

small gap

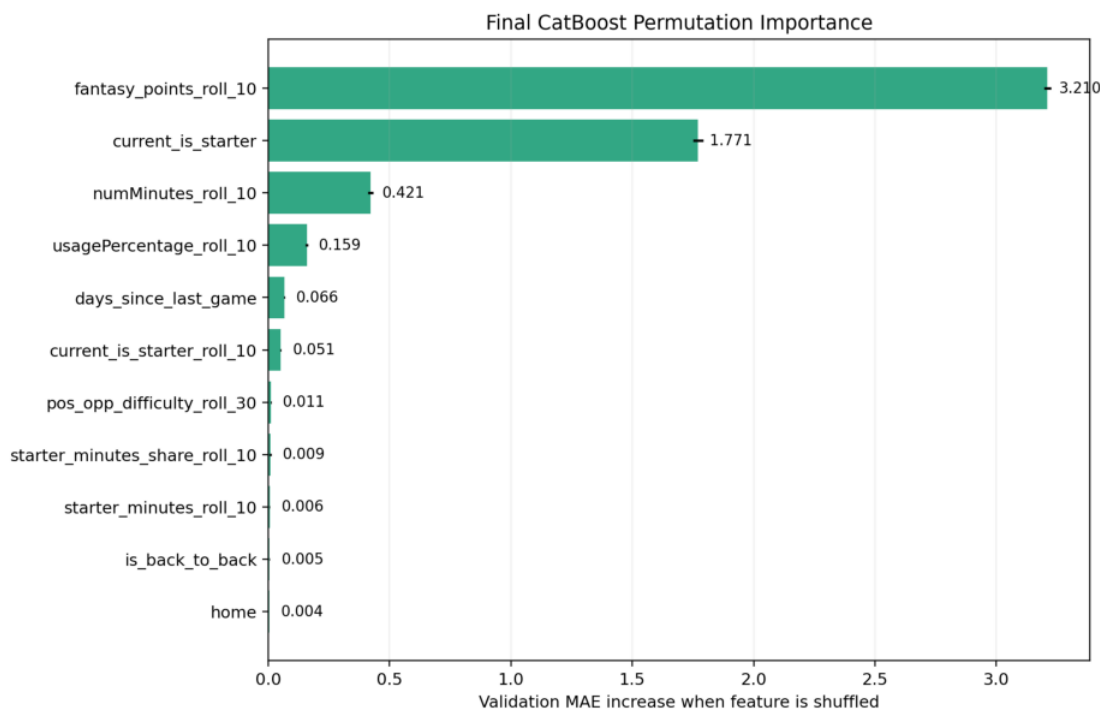
Final Model And Interpretation

The selected final model was evaluated once on the untouched 2025 season.



Final selected model: catboost using `lineup\_aware\_starter\_flag\_roll10`. This is a lineup-aware model: it is valid once confirmed starter status is available, and it does not use actual current-game minutes.

Permutation importance for final model



Permutation importance measures how much validation MAE increases after shuffling one feature. Recent fantasy production and starter status dominate.

Conclusions, Limitations, And Next Steps

What the project proved and what would improve a production version.

Main finding

A leakage-controlled NBA fantasy-points model can substantially beat simple baselines using recent production, role, usage, and lineup-aware starter status. The final model generalized from validation to the unseen 2025 test season with only a small performance change.

Key conclusions

- ? Current-game minutes are highly predictive but invalid before the game; below-7 MAE results were correctly treated as leakage diagnostics.
- ? The final CatBoost model balanced accuracy, interpretability, and reproducibility better than more complex PCA/Isolation variants.
- ? The strongest legal predictors were recent fantasy points, confirmed starter status, recent minutes, and usage rate.
- ? Expanding time CV showed that the main models were stable across earlier chronological validation windows.

Limitations

- ? The final model is lineup-aware; it needs confirmed starter status. A stricter early pre-lineup model is possible but less accurate.
- ? The project does not yet include live injury reports, betting totals, projected minutes, or late scratches.
- ? PCA and anomaly features improved validation accuracy but reduce explainability, so they are best treated as advanced extensions.

Recommended next steps

- ? Add official injury and lineup feeds.
- ? Add projected minutes from an external source and compare against the leakage diagnostic boundary.
- ? Keep 2025 as the final test reference unless the project is rerun with later seasons.